

Proof. **(1) Set the conjugate weights.** Define

$$\alpha := \frac{1}{p}, \quad \beta := \frac{1}{q}.$$

Since $p, q > 1$, we have $\alpha, \beta \in (0, 1)$ and

$$\alpha + \beta = \frac{1}{p} + \frac{1}{q} = 1.$$

(2) Degenerate cases. If $a = 0$ or $b = 0$, then $ab = 0$ while $\frac{a^p}{p} + \frac{b^q}{q} \geq 0$, so the inequality holds with equality. Hence we may assume $a > 0$ and $b > 0$ in what follows.

(3) Weighted AM–GM. For any non-negative numbers X, Y and weights $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$, the weighted arithmetic–geometric mean inequality states

$$\alpha X + \beta Y \geq X^\alpha Y^\beta. \quad (3)$$

Equality in (3) occurs iff $X = Y$.

(4) Apply (3) with $X = a^p$, $Y = b^q$. Since $a, b \geq 0$, also $X, Y \geq 0$. Substituting gives

$$\alpha a^p + \beta b^q \geq (a^p)^\alpha (b^q)^\beta. \quad (4)$$

(5) Simplify the right-hand side of (4). Using $\alpha = 1/p$ and $\beta = 1/q$,

$$(a^p)^\alpha (b^q)^\beta = (a^p)^{1/p} (b^q)^{1/q} = ab,$$

because the bases are non-negative and we take the principal roots. Inserting this into (4) yields

$$\frac{a^p}{p} + \frac{b^q}{q} \geq ab. \quad (5)$$

(6) Conclusion. Inequality (5) is precisely Young’s inequality. Together with the degenerate case handled in step 2, the statement holds for all non-negative a, b .

(7) Equality case. Equality in the weighted AM–GM occurs exactly when $X = Y$, i.e. when $a^p = b^q$. Together with the trivial possibilities $a = 0$ or $b = 0$, the equality cases are

$$a = 0 \quad \text{or} \quad b = 0 \quad \text{or} \quad a^p = b^q > 0.$$

□